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F23BINFT1M01219

SESSION:

2023-2027

Department of Information Technology

The Islamia University of Bahawalpur

Hospital Management System

CamScanner

In the name of Allah, the most Gracious and the most merciful.

27.10.2026
Ghazala Asghar
BSIT 7th - F23BINFT1M01219
SESSION 2023-2027

Hospital Management System

SOFTWARE REQUIREMENTS SPECIFICATION DOCUMENT

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1. Introduction

The main reason to develop our hospital management system is to make hospital task easy and is to develop software that replaces the manual hospital system into automated hospital management system. This document severs as the unambiguous guide for the developers of this software system.

- The software is for the automation of the Hospital Management.
- it maintains two levels of users:

  _Administrator Level

  _User Level
- The software includes:-

  _ Maintaining Patients details.

  _Providing Prescriptions, Treatments and Advices.

  _Admitting a Patient to the Ward if it is required.

  _Billing and Report generation

1.1 Purpose

If the SRS is written in a well manner it can serve the following purposes. SRS is the agreement between the client and the software developer.

Feedback to the customer- This SRS make sure to the project management stakeholders and client that the development team has actually understood the business requirement documentation properly. This also provides confidence that the team will develop the functionality which has been detailed.

Breaking Down the Requirements- This document is prepared in such a way that break the working into smaller components which makes the participants of the project to understand what is to be done clearly.

The information in this document is organized in such a way that all the developing members in the team will not only understand the boundaries of their work, but also functionality needed to develop and in what order.

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This document make it easy to understand that what order what order the functionality will be developed in means that the developers will have the "big picture"view of the development and easily plan ahead which saves both project time and cost.

Facilitate other Documentation-The SRS forms the basis for other important documents such as the Software Design Specification.

Validate the Project-This document initially helps to validate with the client that the product which is being delivered meets what the client has asked for. It means that the client should be satisfied with the product which we will have as output.

1.2Characteristics

Some Characteristics of a Software Requirement Specification can be as follows:

1.1.1. Accuracy

We will make the accuracy of the software and the data entered to the database.

1.1.2. Consistency

The is consistent from start to end. It helps the readers to understand the requirements well.

1.1.3. Completeness

The SRS contain all the requirement stated in the business requirement specification specified by the user.

1.1.4. Clarity

This SRS clearly state what the user want from the software system.

1.1.5. Modifiability

When there is need, the development team and the user modified this SRS.

1.1.6. Traceability

Each requirement stated in the SRS is uniquely associated to a source such as a use case or interaction documentation etc.

1.1.7. Verifiability

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When to project is developed, the user/client will be able to verify that all the agreed deliverables have in fact been produced meet the specified project management system.

1.3 Scope

The details of the current system can be prescribes as follows:

- The current system is a file based system to handle the hospital process. When a patient arrive he make an appointment at the reception to consult to the doctor. All the patient details are recorded in a file. Then the patient is diagnosed for the symptoms of the related disease and its details, ward details and other necessary details are being recorded in the files. Calculations of the bills and inventories are also done manually.
- For the current file based system it is much difficult for the management of the hospital to secure these files. These files can easily be damaged by insects, fire and natural disasters etc. These files can also be misplaced.
- It is tough and time taking process for the management if we want to check for the previous record of any patient in this type of file management system.
- There is a high tendency of making mistakes when working manually. It is hard to rely on the accuracy of the calculation done manually too. It is more obvious for the problems to arise.
- There is a need of a large storage area to store the files while working with manual based system. An issue may arise while there is a limited storage area.

We plan to overcome the above mentioned issues with the help of a standalone application to manage the major tasks of the hospital system.

- The hospital management system we are going to implement will be covering all basic processes done in the hospital. It would handle the employee and salary management, patient management, theater and ward management, laboratory management, transport management, pharmacy management, OPD management and emergency management.
- In employee and salary management, the system will record the attendance, shifting of employees, their holidays and their EPF/ETP and OT hours. This is more efficient and more reliable because the system avoids incorrect inputs whenever they are occurred.

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- Implementing the patient management, the system will record the details of the patient, ward assigned to the patient, lab reports and calculations and billing information.
- Implementing the theater and ward management, records details of surgeons, in-patients who are assigned for the wards, different wards details and surgery details. The pharmaceuticals used within the theater are managed as well. So the details of surgeons, patients and surgeries are well organized and can be easily accessed whenever needed. Surgery reports, ward progress reports, in-ward patient progress details are generated and history can tracked too.
- The vehicle & transport management system handles all the data on the ambulance transport. It manages the time slots of ambulances, driver's and employee details of transport section and provides bill generatingfacility. It also provides details about the availability of the ambulances.
- The pharmacy stock management system is responsible for proper management of drug stocks, pop ups the notifications of expiry dates of the stock items.
- In OPD and emergency management unit, the manual doctor channeling is automated. So the staff does not need to spend time on writing appointments records and updating them in files. The appointment number issuing becomes easier and efficient. It also make it efficient to keep records of the equipments and medicines related to the emergency treatment unit. The system provides the quick interface to record the details of the patients and this information can be viewed and accessed by any external party.

1.4 Definitions, Acronyms, and Abbreviations.

HMS : Hospital Management System.
SQL : Structured Query Language.
GUI : Graphical User Interface.
SRS : System Requirement Specification.
DFD : Data Flow Diagram.
Report : An account of patients.

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1.5 References

There are following reference used to collect data and to accurately complete the SRS:

- Google(www.google.com)
- Wikipedia(www.wikipedia.com)
- Study the Database Management System
- Discussion with teacher on different topics.

1.5 Overview

Hospital Management System is a process of implementing all the activities of the hospital in a computerized automated way to fasten the performance

This project is to maintain the patient details, lab reports and to calculate the bill of the patient. You can also manually edit any patient details and issue bill receipt to patient within few seconds.

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2. The Overall Description

The basic purpose of my project entitled as "Hospital Management System" is to computerized the Front Office Management of Hospital and the develop a software which is user friendly, simple, fast and cost-effective. It deals with the collection of patient's information, diagnosis details etc. Traditionally, it was done manually.

The main function of my system is to register and store patient details and doctor details and retrieve these details as and when required and also to manipulate these details meaningfully.

2.1 Product Perspective

The old system follow the manual procedures to keep track of its day to day activities. When scenarios such as patient information handling, employee handling, stock handling, financial analysis and report generation is taken into consideration there exist many issues with regard to efficiency, security, accuracy and reliability. Due to improperly managed details medical center faces quite a lot of difficulties in accessing past data as well as managing present data. The manual file system which are being used at present require storage facility which is also another overhead.

The fully functional automated hospital management system which will be developed through this project will eliminate the disadvantages caused by the manual system by improving the reliability, efficiency, and performance. The usage of a database to store patient, employee and stock details etc will accommodate easy access, retrieval, search and manipulation of data. The access limitations provided through access privilege levels will enhance the security of the system. The system will facilitate concurrent access and convenient management of activities of the medical center.

2.1.1 Operations

The Hospital Management System will provide the following operations to satisfy the users of the system.

- The admissions of the patients to the hospital.
- Checkup of the patients from the doctors.
- Lab tests details of the patients.
- Surgeries details of the patients.
- Ward details if the patient is admitted to the ward.
- To maintain the records of the medicines given to the patients.
- To calculate the total bills of the patients.

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2.1.2 Site Adaptation Requirements

The current being developed Hospital Management System is an application based system which is developed using Microsoft Visual Studio. It uses Microsoft SQL Server to store data as Database.

2.2 Product Functions

The productwhich is being developed will actually perform the following functions.

OPD and Consultation Management

- Recording patient details
- Issuing numbers according to doctor channeled
- Updating the record with medical prescription
- Printing bill of doctor charges

Employee and Salary Management System

- Attendance and schedule of employees
- Holiday approvals
- Schedule of consulting doctor
- EPF/ETP and OT calculation
- Shares of consulting doctor
- Shares of doctors who has sent patient for other tests

Theater and Ward Management

- Details of surgeon
- Surgery success/failure details
- Complications of the patient, patient history and other details
- Bill calculations and other reports

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Vehicle and Transport Management

- Ambulance Management
- Vehicle deployment details
- Time management
- Drivers and employee details
- Billing calculations

Emergency Treatment and Equipment Management

- Emergency patient details
- Issue tests reports
- Billing calculations
- Lab equipment stock management

Pharmacy Stock Management

- Drug stock management
- Expiry notification
- Billing calculations
- Searching

Lab Management

- Record sample collection details
- Lab resources management
- Lab report generation
- Bill Calculation
- Lab equipment stock management

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2.3 User Characteristics

- **Admin**

  Admin has the full access to the system which means he is able to manage any activity with regard to the system. He is the highest privileged user who can access to the system.

  Key Functions:

- Manage employees, patients and equipments
- Allocate resources
- Administer the changes
- Generate reports
- Manage ambulances
- Manage doctors
- Manage Salaries

- **Employee**

  Interacts with the system most often to supply service to customers.

  Key functions:

  - Keep track of patient details
  - Keep track of test details
  - Make ambulances reservations
  - Keep track of progress of patients
  - Maintain bill details
  - Manage inventory

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2.4 General Constraints

- System is wirelessly networked with an encryption.
- System is only accessible with in the hospital premises only.
- Database is password protected.
- Should use less RAM and processing power.
- Each user should have individual ID and password.
- Only administrator can access the whole system.

2.5 Assumptions and Dependencies

- Each user must have a valid user id and password.
- Server must be running for the system to function
- Users must log in to the system to access any record.
- Only the administrator can delete the records.

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3. Specific Requirements

The current system which is going to be developed is a computer based system for Hospital Management System which was handled manually before. There are following necessary requirements for the system to get developed.

3.1 External Interface Requirements

3.1.1 System Interfaces

System interface is a view from where the user can interact with the system to give input to the user and get his required output from the system. The current being discussed Hospital Management System will provide different GUI to interact the system and to maintain the records of the patients in the hospital.

The system will provide the interfaces which will perform the following functions.

- Patient's Data Entering System

  This system will allow the receptionists to enter the date of different patients being entered in the hospital.

- Patient's Data Maintaining System

  This system will allow the user of system to maintain(update, delete etc) the data of the patient. Only the admin can maintain the data of the patients.

- Wards &Surgeries Management System

  This system will allow the user to manage the records of the ward if the patient is admitted to the ward surgeries on the patientsand details of the surgeons.

- Lab Test Management System

  This system will allow the user to manage the records of the lab tests of the patients.

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- Pharmacy Management System

This system will allow the user to store the details of medicines prescribes to the patients.

- Billing System

This system will calculate the total bills of the patients related to check up, tests, surgeries and medicines etc.

3.1.2 Interfaces

Specific Interface:

The HMS will take the input, process that input and will provide the output.

- Input:

The HMS will take the records of the patients as basic input.

- Processing:

The HMS will process over the records of the patients checkups, their lab tests, their surgeries and medicines assigned to them.

- Output:

The HMS will finally gave the total charges of the treatment of the patients as total bill.

3.1.3 Hardware Interfaces

There are following Hardware Interfaces for the current developing system.

- Laptop/PC

RAM	:	2Gb OR more.
ROM	:	Minimum 512Gb of Hard Disk.

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Processor : Pentium IV (1.2 GHz)

Purpose of this pc is to give information when Patients ask information about doctors,medicine available lab tests etc. To perform such Action it need very efficient computerotherwise due to that reason patients have to wait for a long time to get what they ask for.

- Display Unit (LED/LCD Monitor/TV)

This unit is for display the channel number when the patients cometo see their consultants. It will avoid chaos.And also displayHospital welcome screen, video, information etc.

- Laser Printer (B/W)

Simply this device is for printing bills and view reports.

3.1.4 Software Interfaces

There must be following software specifications in the system to develop and run the current system.

Operating System	:	Microsoft windows 7 OR more.
Front End Software	:	Microsoft Visual Studio.
Database	:	Microsoft Office Access.

3.1.5 Communications Interfaces

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This system is developed to update the manual based Hospital Management System into a computer application based system which is easier to use and maintain. It is easy to store the records of the patients and other details related to the patient in a hospital and also convenient to update those records. this system will work with the patients to keep their records, to keep the records about their diseases, wards information, surgery information and their medicine information etc.

3.2 Functional Requirements

The HMS will have to satisfy the following functional requirements of the user:

User Satisfaction

The system is search that it stand up to the user expectations.

Response Time

The response time of all operation is excellent.

Error Handling

The error handling ability make the system user friendly.

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3.3 Use Case

The Use Case Diagram for the system is as follows.

3.3.1 Use Case Diagram

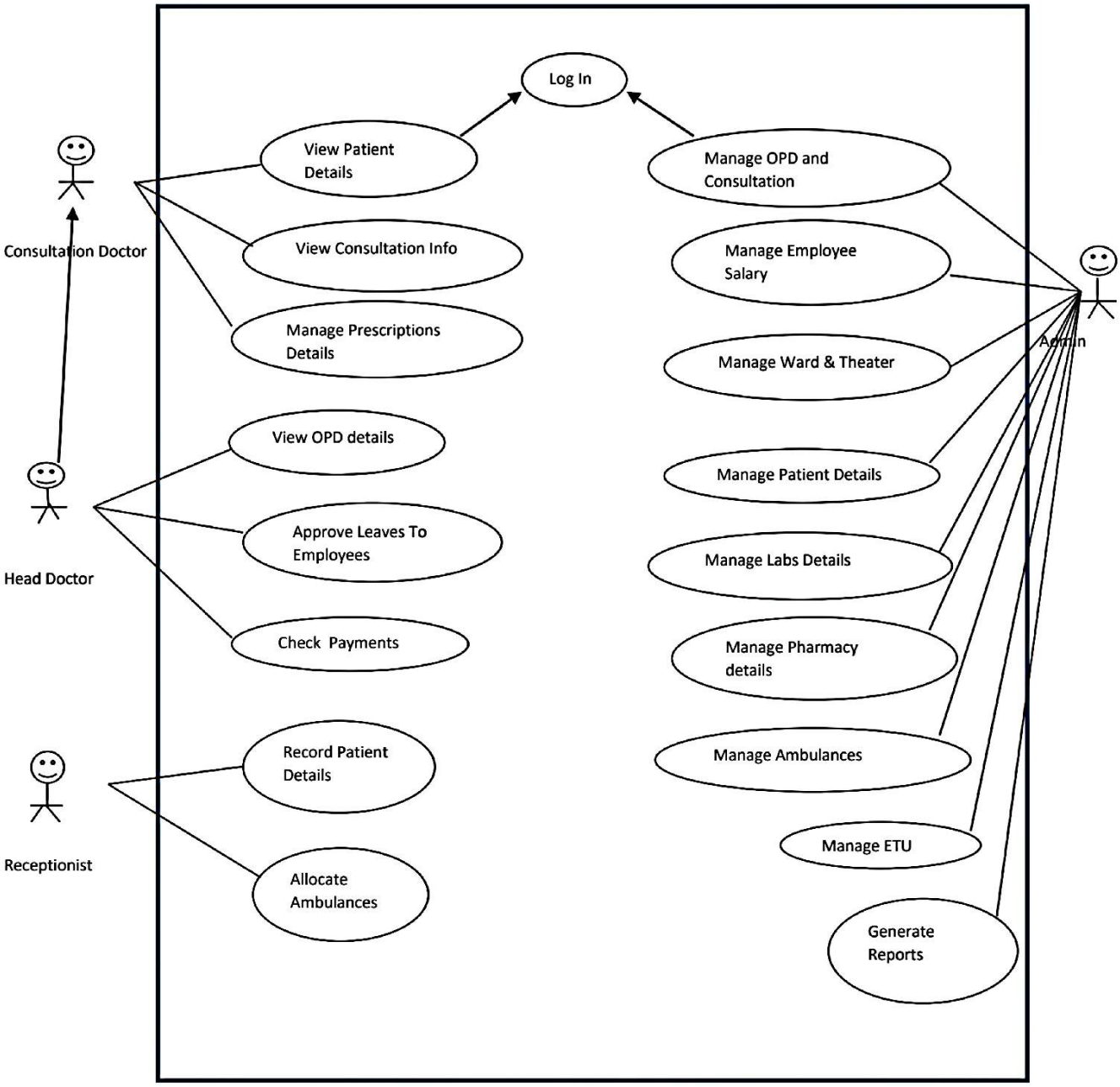


Fig:3.1 Use Case

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3.4 Classes / Objects

There are following classes being used in the current developing project.

3.4.1 Admin class

3.4.1.1 Attributes

admin_ id, admin_ name, admin_ password etc are the attributes of the admin class.

3.4.1.2 Functions

Admin will maintain the records of all the patients, doctors and all information related with the hospital.

3.4.2 Receptionists class

3.4.2.1 Attributes

Receptionists_ id, Receptionists _name, Receptionists _password etc are the attributes of the Receptionists class.

3.4.2.2 Functions

Receptionists will enter the data of the patients in the system.

3.4.3 Patient class

3.4.3.1 Attributes

Patient_ id,Patient_ name,Patient_ password etc are the attributes of the Patient class.

3.4.3.2 Functions

Patients enter the hospital for the treatment.

3.4.4 Doctor class

3.4.4.1 Attributes

Doctor_ id,Doctor_ name,Doctor_ passwordetc are the attributes of the Patient class.

3.4.4.2 Functions

Doctor treat the patients in the hospital for their diseases.

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3.5 Non-Functional Requirements

The system must satisfy Non-functional requirements for the following attributes.

3.5.1 Performance Requirement

Response time-The system will give responses within 1 second after checking the patientinformation and other information.

Capacity-The system must support 1000 people at a time.

User interface- User interface screen will response within 5 seconds.

Conformity -The system must conform to the Microsoft accessibility

3.5.2 Security Requirement

All the administrative and data entry operators have unique logins so system canunderstand who is login in to system right now no intruders allowed except systemadministrative nobody cannot change record and valuable data.

3.5.3Safety Requirements

If there is extensive damage to a wide portion of the database due to catastrophic failure,such as a disk crash, the recovery method restores a past copy of the database that was backed upto archival storage and reconstructs a more current state by reapplying or redoing the operationsof committed transactions from the backed up log, up to the time of failure.

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3.5.4Software Quality Attributes

AVAILABILITY: The system shall be available all the time.

CORRECTNESS: A bug free software which fulfill the correct need/requirements of theclient.

MAINTAINABILITY: The ability to maintain modify information and update fixproblems of the system

USABILITY: software can be used again and again without distortion.

ACCESSIBILITY: Administrator and many other users can access the system but theaccess level is controlled for each user according to their work scope.

ACCURACY: The reliability on the information/output. Can depend/be sure of theoutcome.

STABILITY: The system outcome/output won't change time to time. Same output willbe given always for a given input

3.5.5Business Rules

Want take the responsibility of failures due to hardware malfunctioning.

Warranty period of maintaining the software would be one year.

Additional payments will be analyzed and charged for further maintenance

If any error occur due to a user's improper use. Warranty will not be allocated to it.

No money back returns for the software.

Trust bond placement should be done before designing and coding. An advance or anAgreement.

3.5.3 Availability Requirement

The system should be available all the time. User friendly system which is in access of people around the world should work 24 hours.

3.5.6 Maintainability Requirement

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The system should be easy to maintainable. It should be easier for the system developer to easily make changes in the system if required.

3.6 Logical Database Requirements

It represent the logically representation of data that is stored in the database some of them are described below:

- It should be secure and password security.
- It has facility to remove the error.
- The data should not be duplicate.
- Fast query processing in the system.
- It maintain the records the efficiently.
- It defines the logical structure of database that data is stored logically in the database.

3.7 Design Constraints

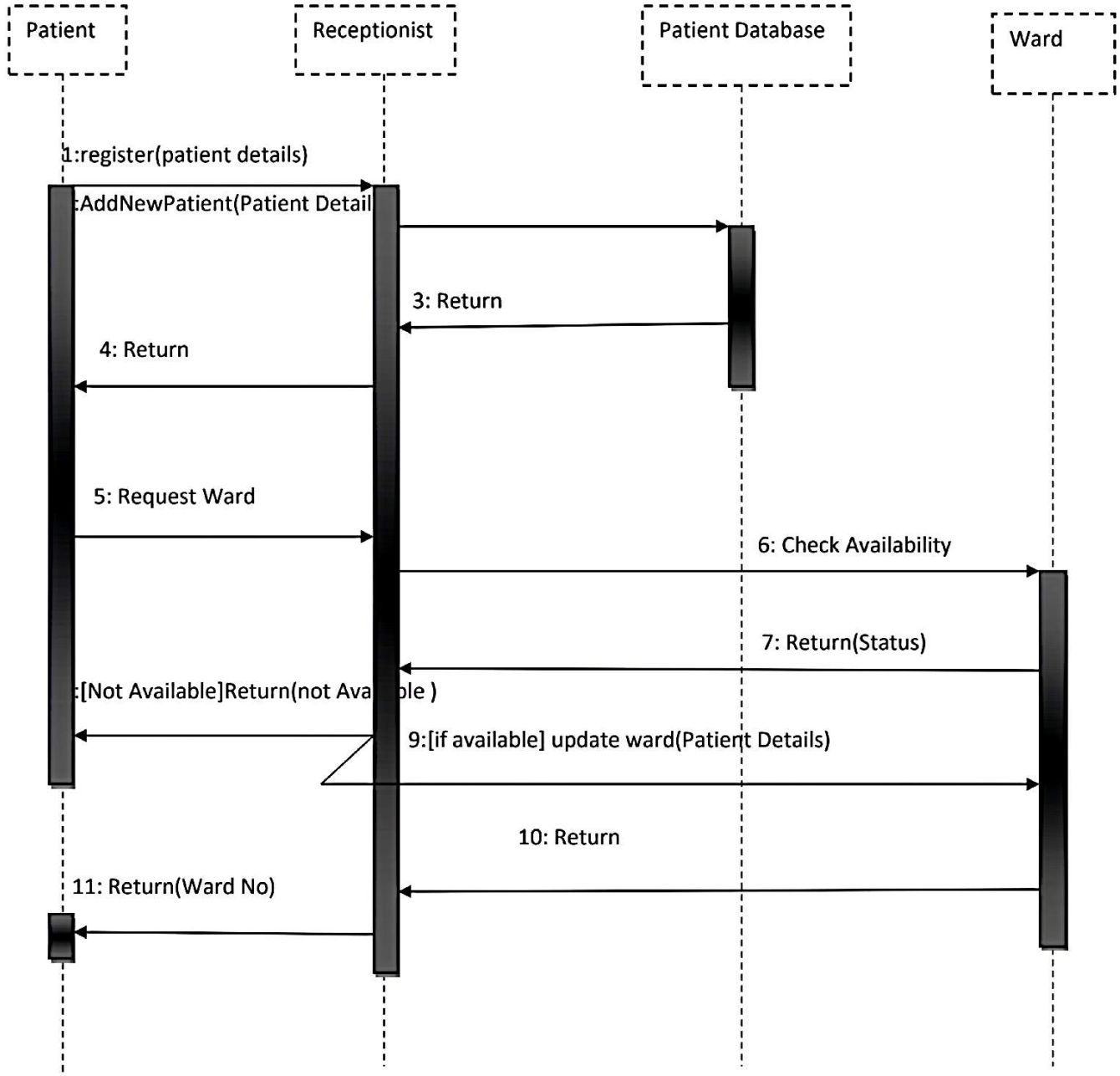
This will help the doctors or users to view the records ofthepatients immediately whenever necessary. They can also calculate the bill of theparticular patients. This software also has the ability to add, update anddelete the record whenever needed. This project will help to smoother theprocess of the hospital activities.

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4. Analysis Models

List all analysis models used in developing specific requirements previously given in this SRS. Each model should include an introduction and a narrative description. Furthermore, each model should be traceable the SRS's requirements.

4.1 Sequence Diagrams



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4.2 Data Flow Diagrams (DFD)

4.2.1 Level -0

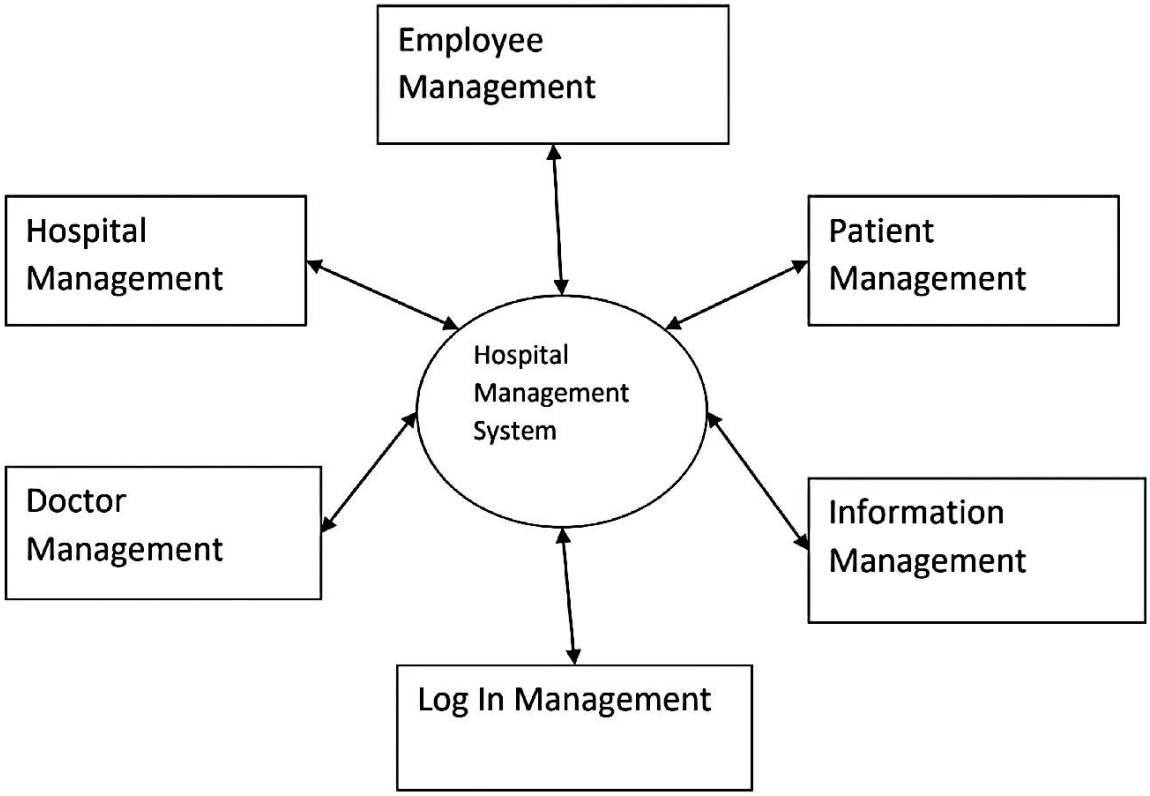
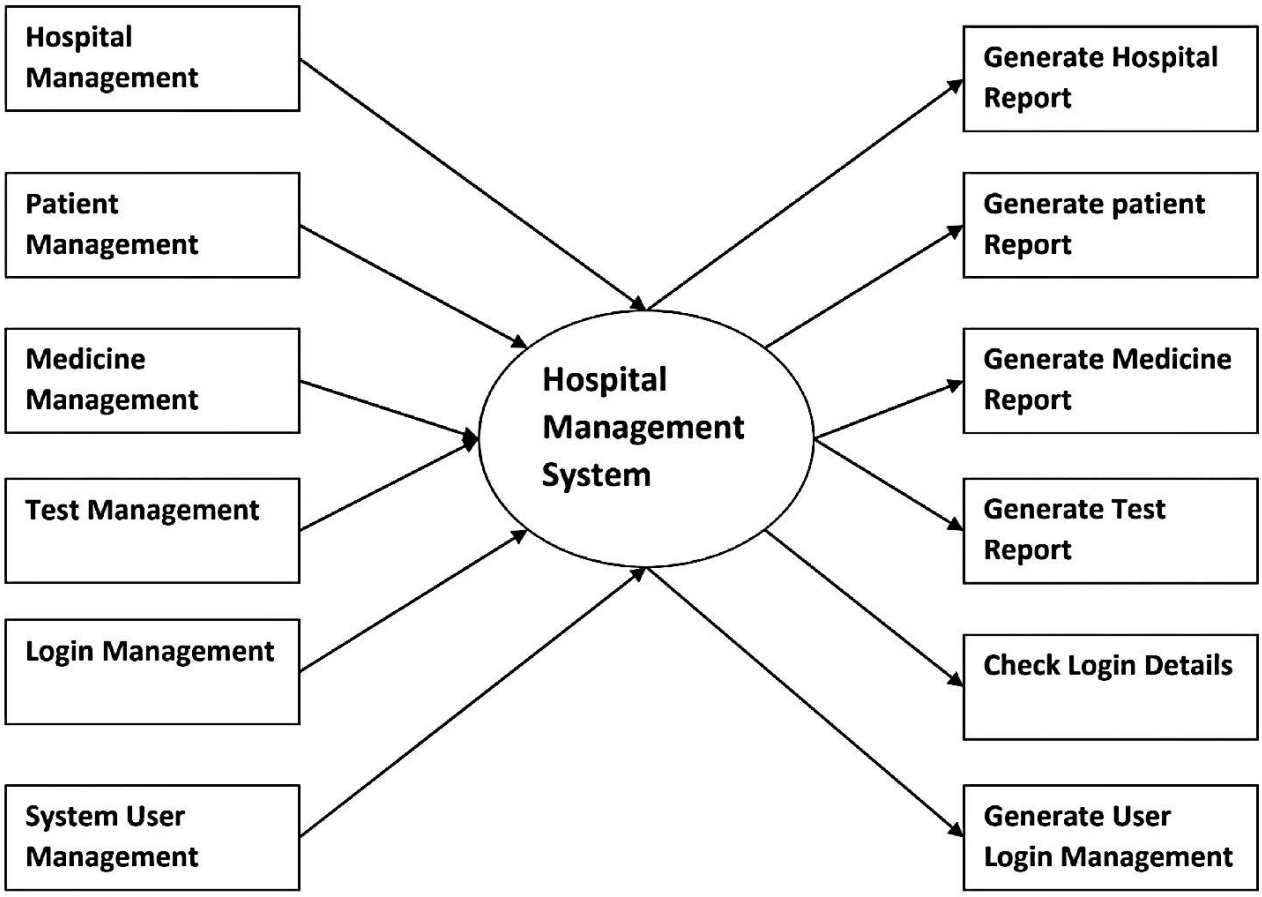


Fig:4.2 Level-0 DFD-Hospital Management

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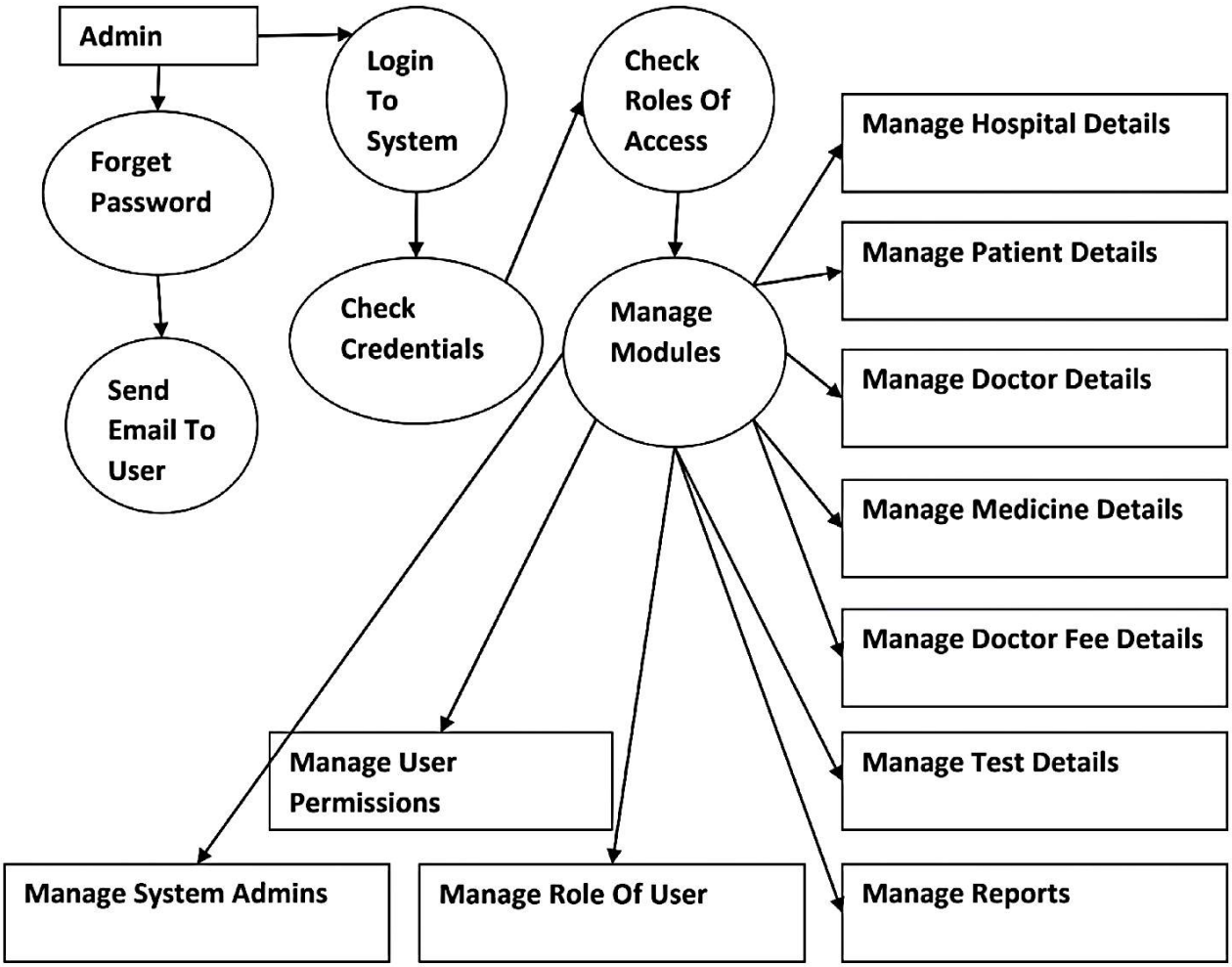
4.2.1 Level-1



4.3 Level-1 DFD-Hospital Management

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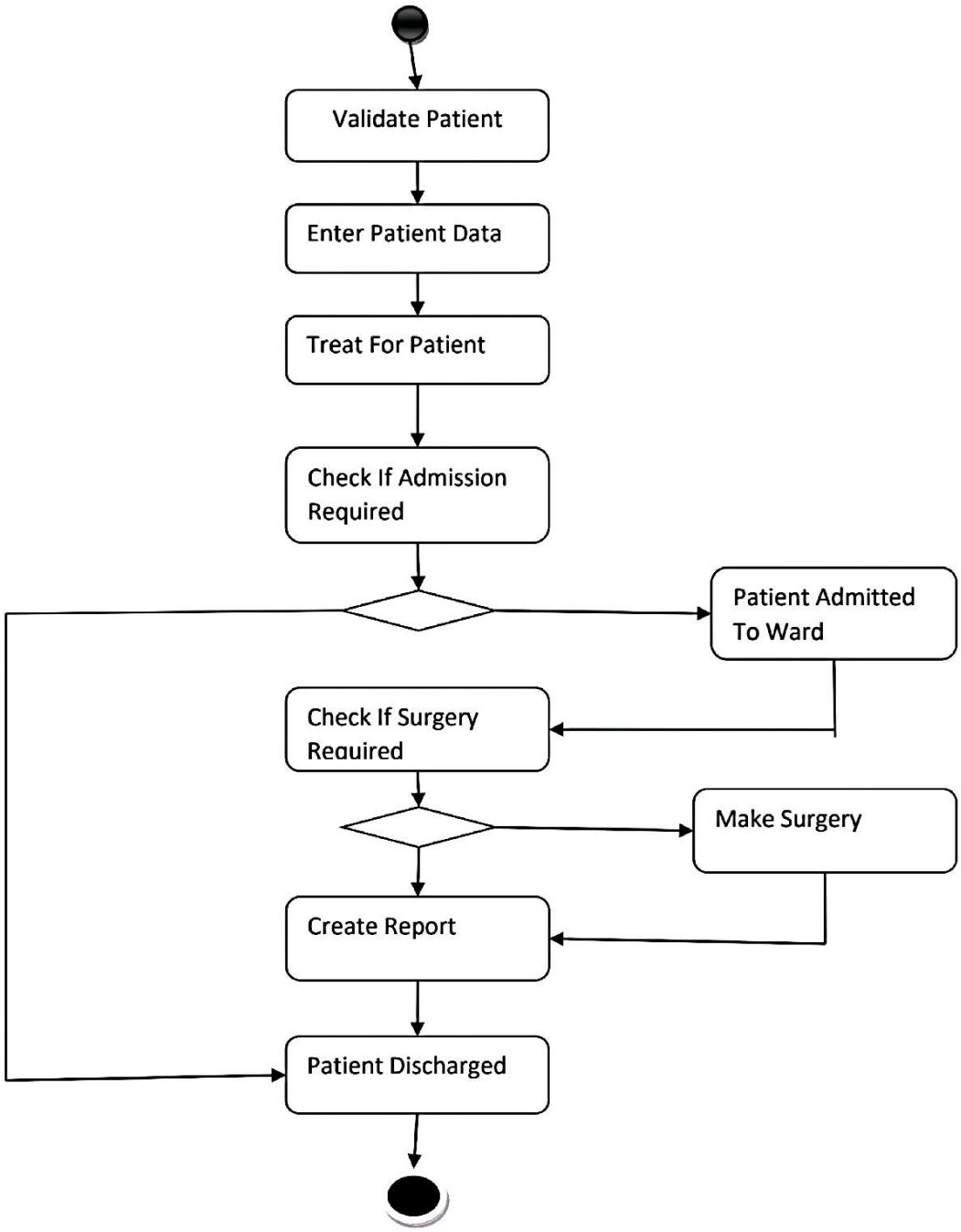
4.2.1 Level-2



4.4 Level-2 DFD-Hospital

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4.3 State-Transition Diagram



4.5 State-Transition Diagram

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4.4 Conclusions

This SRS document is used to give details regarding Hospital Patient Info Management System. In this all the functional and non-functional requirements are specified inorder to get a clear cut idea to develop a project.

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